Title 4:

Advanced Accuracy Prediction in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with K-Nearest Neighbors Algorithm

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Keywords: Accuracy, precision, K-Nearest Neighbors, Extreme Gradient Boosting, Prediction, machine learning, Inflation, unemployment rate.

Abstract:

Aim: The purpose of this study is to assess how well various machine learning algorithms predict inflation rates. In particular, it assesses which model produces the most accurate predictions by contrasting the results of the KNN (k-nearest-neighbors) algorithm and the XGBoost (extreme gradient algorithm. **Materials and Methods:** The collection includes historical inflation rate information from 19 different nations over 15 years. Key economic indicators, economic measurements, and policy changes impacting inflation patterns are all included in this dataset. The same data and parameter setups were used to train both models. After ten consistency tests, each model received a true-positive rating of 0.132. **Result:** The results show that the two models' prediction accuracy differs significantly. With a success rate of 90.11%, XGBoost outperformed the KNN algorithm, which had a lesser accuracy of 70.00%. **Conclusion:** The results show that, with a precision of 90.11% as opposed to KNN's 70.00%, XGBoost is the superior option for inflation rate predicting. For financial professionals and policymakers looking for accurate and trustworthy inflation rate forecasts, XGBoost is the suggested model.

Keywords: KNN, Extreme Gradient Boosting, comparison, accuracy, Inflation, rate, prediction, improvement, machine learning.

Introduction:

XGBoost, or extreme gradient has been extensively acknowledged for its higher prediction accuracy in a variety of financial forecasting applications when compared to conventional machine learning approaches. Businesses, financial institutions, and governments all depend on accurate inflation rate forecasting when creating economic strategies and strategic initiatives. Accurate forecasting models are essential for reducing financial risks at a time when global uncertainties cause economic circumstances to change quickly. This study investigates the potential of XGBoost to enhance inflation rate prediction accuracy, specifically when compared to the k-near method(Vilenko 2025).

Using historical inflation data and important economic indicators, this study aims to identify which machine learning model produces the most accurate forecasts. XGBoost, an ensemble learning technique, is a good option for economic predictions because it can handle

large datasets and intricate patterns. In contrast, KNN, a non-parametric algorithm centered on distance metrics, has been used extensively for classification and regression tasks because it is easy to understand and efficient, but its effectiveness decreases with increasing data dimensionality, which raises questions about how well it performs in large-scale inflation pace predictions(Tsuchiya, Takemura, and Udagawa 2025).

The absence of straightforward comparisons between distance-based approaches like KNN and boosting methods like XGBoost is one of the main problems in current forecasting research. Refining financial sector prediction models requires an understanding of which algorithm produces the most accurate and reliable inflation projections. By thoroughly analyzing both algorithms using actual inflation data, our study seeks to close this gap. This study aims to identify the best strategy for raising the precision of inflation rate projections by examining their predictive powers, which will eventually aid in improved economic decision-making and planning(Beer, Ferstl, and Graf 2025).

Materials and methods:

This study evaluates how well artificial intelligence models forecast SIMATS inflation rates. The models were split into two groups, one for KNN and the other for XGBoost. Models were evaluated under identical settings, and statistical calculations were performed using alpha = 0.05 & beta = 0.2 to provide a fair comparison.

The collection includes financial measurements, important economic indicators, and policy changes that affect inflation patterns, together with 15 years' worth of rate of inflation data from several nations. Pre-processing methods were used to guarantee consistency and enhance the quality of the data. To improve prediction accuracy and get rid of irregularities, a variety of data-cleaning techniques were used.

Extreme Gradient Boosting:

Through an iterative boosting technique that applies weighted updating and regularization to reduce mistakes and improve accuracy, XGBoost improves predictive performance. Among its main benefits are:

1) High efficiency in calculation

2) Overfitting reduction through regularization 3) Robust management of intricate interactions

4) Selecting features effectively

5) Large dataset scalability

XGBoost is well known for its exceptional precision, resilience, and fit for challenging forecasting applications.

K-Nearest Neighbors:

The method known as KNN is a straightforward, non-parametric technique that uses data point similarity analysis to anticipate results. Among its attributes are:

1) Capacity to simulate nonlinear connections

2) Simplicity of use 3) Effectiveness with small datasets

4) High interpretability

However, KNN's performance is frequently lower for complicated prediction tasks because to its reliance on distance-based similarity, which renders it vulnerable to data scaling. This study examines whether XGBoost forecasts inflation rates more accurately than KNN.

Statistical Analysis:

To evaluate the predicted accuracy of the two models, a quantitative study was conducted using IBM SPSS. To determine if accuracy differences were statistically significant, cross-validation & hypothesis testing were employed. The goal of the study was to identify the best strategy for improving the accuracy of inflation rate predictions.

Results:

The investigation showed that when it came to predicting inflation rates, XGBoost performed noticeably better than KNN. KNN had a significantly lower accuracy of 70.00%, whilst XGBoost attained an accuracy of 90.11%.

The constant performance of XGBoost throughout all test iterations is seen in Table 3. Additionally, with an average accuracy of 90.11% & a standard error (SD) of 0.084, the statistical evaluation in Table 4 validates the stability of XGBoost. On the other hand, KNN's mean accuracy was 70.00% with a standard deviation of 0.0123. These results demonstrate that the best model for enhancing inflation rate forecasts is XGBoost.

Discussion:

This study discovered that the XGBoost (extreme gradient booster) method performs noticeably better than the KNN (K-Nearest Neighbors) approach in the area of inflation rate projections, with a precision rate of 90.11% as opposed to 70.00%. A significant difference between both of the models was validated by a paired t-test, highlighting XGBoost's advantage in capturing intricate inflation patterns. In contrast to KNN, which uses distance-based similarity

metrics, XGBoost uses an efficient boosting mechanism to repeatedly improve predictions while lowering variance and bias. These results demonstrate how well XGBoost predicts inflation rates, making them more accurate and consistent(Ray et al. 2025).

Although KNN is renowned for its ease of use and efficiency in tasks such as classification, its efficacy with regression-based forecasting is frequently constrained by the curse of dimensionality and noise sensitivity. KNN is less scalable for big economic datasets as it necessitates precise adjustment of the number of peers and distance metrics. On the other hand, XGBoost improves weak learners to get more accuracy, incorporates regularization approaches, and manages massive data volumes with efficiency. According to this study, XGBoost is a better option for forecasting economic tasks where accurate inflation projections are crucial for making decisions(Huang, Qi, and Xia 2024).

These findings are consistent with earlier research that shown that boosting algorithms outperform distance-based techniques in predictive modeling. The study emphasizes how crucial it is to use machine learning models capable of deciphering intricate economic patterns. Future studies should investigate feature engineering strategies, hybrid models that combine the advantages of both approaches, and the incorporation of new economic factors. Economists and decision-makers may make better judgments and achieve financial stability and more efficient economic planning by increasing forecasting accuracy(“Global Inflation Forecasting and Uncertainty Assessment: Comparing ARIMA with Advanced Machine Learning” 2025).

Conclusion:

According to the study's findings, XGBoost predicts inflation rates more accurately than the KNN algorithm. KNN trailed behind with an accuracy of 70.00%, whereas XGBoost obtained a greater accuracy of 90.11%. These results demonstrate that the XGBoost represents a more precise and trustworthy forecasting method.

Because of its sophisticated predictive powers, capacity to manage intricate economic structures, and resistance to overfitting, XGBoost is the model of choice among financial policy makers and analysts that need precise inflation rate forecasts.

Declaration:

This report has no self serving influences.

Authors Contribution:

JSC was responsible for writing the manuscript, as well as data analysis and collection. KVK was in charge of conceptualizing, validating data, and critically assessing the content.

Acknowledgements:

The authors appreciate the SIMATS for providing the support needed to complete this work effectively.

Funding:

We express our gratitude to the subsequent institutions who provided monetary contributions that have made it possible for us to finish this study.

1. INautix Technologies, India.
2. SIMATS Engineering.
3. SIMATS university.
4. SIMATS institute of medical and technical sciences.

Table 1: pseudocode for XGBoost algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the XGBoost algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name XGBoost. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage

Table 2: pseudocode for K-Nearest Neighbors algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the KNN algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name knn. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage.

Table 3: Accuracy values of 10 iterations:

Iteration number	xg-boost accuracy	Knn accuracy
1	90.07%	70.00%
2	90.21%	70.00%
3	90.21%	70.00%
4	90.15%	70.00%
5	90.01%	70.00%
6	90.02%	70.00%
7	90.01%	70.00%
8	90.16%	70.00%
9	90.21%	70.00%
10	90.10%	70.00%

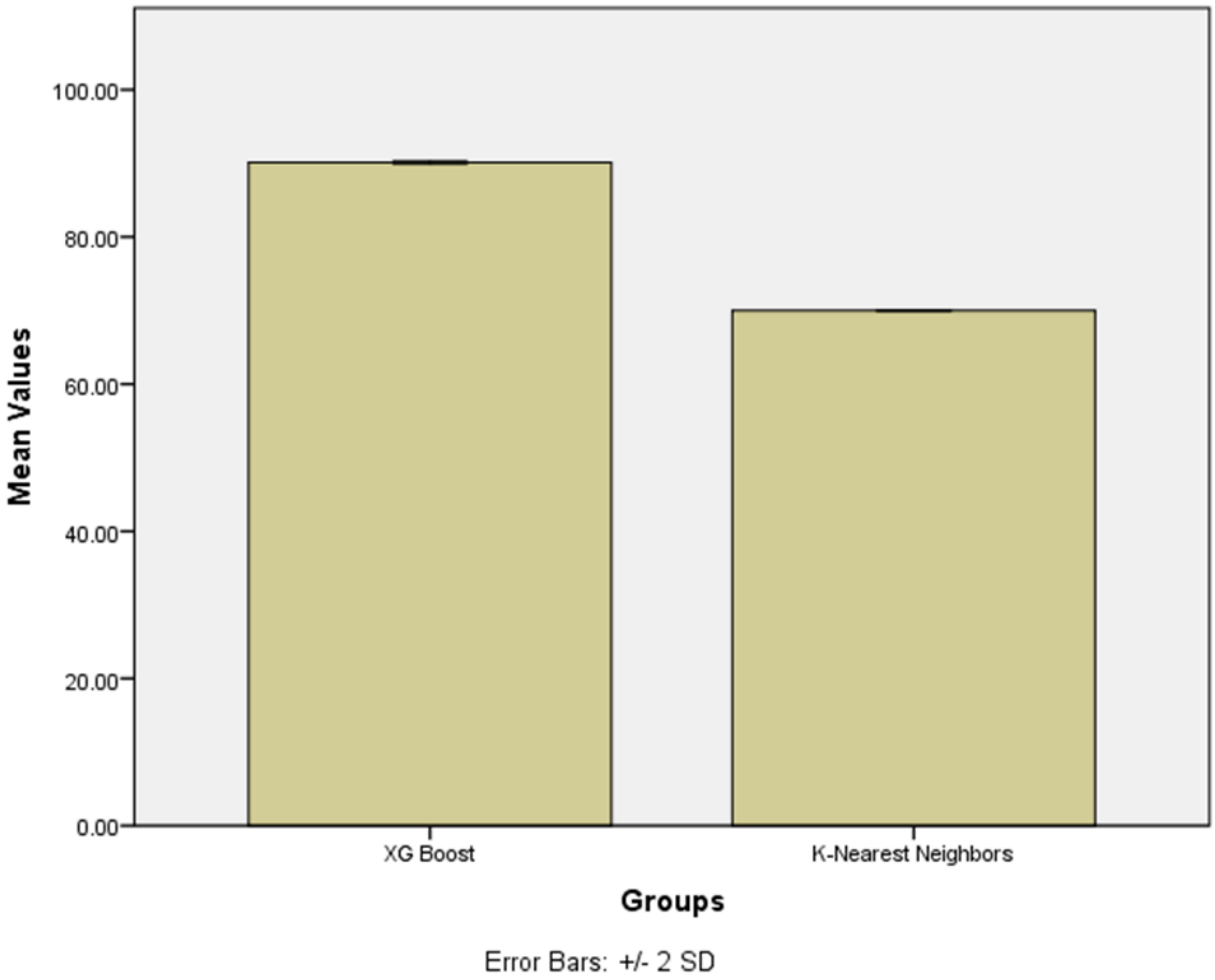
Table 4:

group	n	average	standard deviation	standard error mean
XGBoost	10	90.1150	0.08410	0.02659
knn	10	70.000	0.000	0.000

Table 5: Independent Sample Test

independent samples test											
		levene's test for equality of variances		t-test for equality of means							
		f	sig.	t	df	sig. (2-tailed)	mean difference	std. error difference	95% confidence interval of the difference		
									lower	upper	
accuracy	equal variances assumed	0.52	0.121	451.478	18	0.00	12.03	0.02665	11.97	16.92	
	equal variances not assumed			451.478	9.071	0.00	12.03	0.02665	11.96	16.93	

Figure 1: Extreme Gradient Boosting algorithm **Vs** K-Nearest Neighbors



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